

Chapter 6 Annual Worth Analysis

- **Introduction**

- Annual worth (AW) analysis is a variant of the present worth analysis discussed in Chapter 5.
- However, AW analysis has many advantages that make it a useful technique for comparing alternatives.

- **Advantages of AW analysis**

- It's a popular analysis technique.
- It's easy to understand. Results are reported in \$/year.
- It simplifies comparing alternatives when cash flows repeat. No need to compare the alternatives over the LCM. Compare over life cycle of each alternative.

- **How does it work?**

- For alternative j , find the uniform annual series, with value AW_j , which is equivalent to all the cash flows of the alternative at the decision maker's MARR.
- An alternative j with $AW_j \geq 0$ is economically viable.
- Compare annualized series (the AW_j s) of all alternatives
- The alternative with largest AW_j is selected.
- When cash flows repeat, AW_j is found over the duration of Alternative j . No need to compare over the LCM of lives.
- If cash flows don't repeat, AW_j is found over a study period.

- **Keep in mind**

- PW and AW analysis are equivalent
- An alternative has $AW \geq 0$ if and only if $PW \geq 0$.
- An alternative has largest AW among a set of alternatives if and only if it has the largest PW.

- **Capital Recovery (CR) calculation**

- Capital Recovery (CR) is the annualized equivalent of the initial investment P and salvage value S of an alternative,

$$CR = -P(A/P, i, n) + S(A/F, i, n) .$$

- Commonly, CR is added to the annual operating costs (AOC) to get AW,

$$AW = CR + AOC .$$

- **Annual worth analysis of permanent investments ($n = \infty$)**

- This is similar to the capitalized cost analysis in Chapter 5.
- For a cash flow R , recurring every n_R years, starting Year n_R ,

$$A_R = R(A/F, i, n_R) = R \left[\frac{i}{(1+i)^{n_R} - 1} \right] .$$

- For a non-recurrent cash flow C , occurring at Year n_C ,

$$A_C = C(P/F, i, n_C)(A/P, i, \infty) = \frac{Ci}{(1+i)^{n_C}} .$$